Line (GAC): F ROAD STRUCTURES

Competency: F2 Verify slopes, crowns and grades

Objectives

To be competent in this area, the individual must be able to understand how to verify basic road grade, crown and slope.

LEARNING TASKS	CONTENT
1. Confirm plans/blueprints	• Metric and Imperial measurement • Scale • Abbreviations, symbols common to civil drawings • Distinguish horizontal distance from slope distance
2. Define role of Surveyor	• Determine cuts and fills of road construction at specified centerline points on profile • Determine depth of pipe at specified points • Calculate the volume of aggregate types allowing for settlement i.e. road base or pipe bedding • Create material lists i.e. pipe, valves, etc.
3. Define role of Grade Person	• Transfers info to offset stakes
4. Perform typical calculations	• Use 'real world' examples of plan sets to describe drawings • Plan view: drawing with the view from overhead • Profile view: drawing of side view with the vertical scale exaggerated, usually 10x the horizontal scale; showing original ground and design grades of roads, pipe, manholes, culverts, bridges; cuts and fills shown are at the centerline only; often on the same page as a plan view aligned on the same horizontal scale • Section: drawing of a 'slice' of works at right angle to the centerline with the same horizontal and vertical scale • Additional features of drawings: scale, north orientation, legends, creation and revision dates, "created by" info, detail drawing insertions

Achievement Criteria

Performance	Learner will demonstrate basic knowledge and use of various grade-measuring instruments. Learner will demonstrate safe practices during practical training and on the job.
Conditions	The learner will demonstrate the use of: • Spirit levels • Builders levels • Laser level • Automatic slope control
Criteria	The learner will maintain specified grade and slope.

Line (GAC): F ROAD STRUCTURES

Competency: F3 Describe final grade preparation

Objectives

To be competent in this area, the individual must be able to understand final grade preparations to road structures.

LEARNING TASKS	CONTENT
1. Describe how to check and raise utilities	• Describe methods of raising castings and related traffic control issues
2. Describe fine grading	• Discuss fine grading, soil compaction and surface preparation
3. Describe boning crown/cross-fall	• Ensure crown and cross-fall specifications are understood and proper calibrations made to equipment
4. Describe limit of asphalt layout	• Determine width of finished mat

Line (GAC): G END PRODUCT SPECIFICATIONS

Competency: G1 Describe gradation

Objectives

To be competent in this area, the individual must be able to understand the quality control plan as it relates to MOT and the general ministry audit process.

LEARNING TASKS	CONTENT
1. Describe MOT Specs	• Review Part A General including definitions • Preparing a Quality Control Plan for evaluation before commencing the Work and providing at the production site a testing facility to provide the data needed to implement that plan • Supplying, screening, crushing, processing and improving aggregate to produce asphalt mix aggregate • Supplying and delivering asphalt cement and spray primer meeting the requirements of Section 952 • Preparing mix designs which, once verified by the Ministry Representative, become the basis for the accepted Job Mix Formula • Heating the asphalt mix aggregate and mixing it with asphalt cement to produce asphalt mix that meets the Job Mix Formula • Hauling, placing, compacting and finishing the asphalt mix
2. Describe the audit process	• Quality control plan requirements
3. Describe penalty/bonus	• Ensure crown and cross-fall specifications are understood and proper calibrations made to equipment
4. Describe limit of asphalt layout	• Discuss payment adjustment form Part B of section 502 • Review Table 502-G

Line (GAC): G END PRODUCT SPECIFICATIONS

Competency: G2 Describe oil content

Objectives

To be competent in this area, the individual must be able to understand Section 502 of the MOT Design Build Specifications for Highway Construction regulations as it relates to oil content.

LEARNING TASKS	CONTENT
1. Define Section 502	• Oil content requirements

Line (GAC): G END PRODUCT SPECIFICATIONS

Competency: G3 Describe application

Objectives

To be competent in this area, the individual must be able to understand Section 502 of the MOT Design Build Specifications for Highway Construction regulations as it relates to asphalt applications.

LEARNING TASKS	CONTENT
1. Define Section 502	• Asphalt application standards

Line (GAC): G END PRODUCT SPECIFICATIONS

Competency: G4 Describe segregation

Objectives

To be competent in this area, the individual must be able to understand Section 502 of the MOT Design Build Specifications for Highway Construction regulations as it relates to segregation of product.

LEARNING TASKS	CONTENT
1. Define Section 502	• Segregation of asphalt

Line (GAC): G END PRODUCT SPECIFICATIONS

Competency: G5 Describe density

Objectives

To be competent in this area, the individual must be able to understand Section 502 of the MOT Design Build Specifications for Highway Construction regulations as it relates density.

LEARNING TASKS	CONTENT
1. Define Section 502	• Density specifications

Line (GAC): G END PRODUCT SPECIFICATIONS

Competency: G6 Describe smoothness

Objectives

To be competent in this area, the individual must be able to understand Section 502 of the MOT Design Build Specifications for Highway Construction regulations as it relates asphalt smoothness.

LEARNING TASKS	CONTENT
1. Define Section 502	• Smoothness standards

Line (GAC): H OPERATE MILLING AND RECLAIMING EQUIPMENT

Competency: H1 Describe milling and reclaiming equipment

Objectives

To be competent in this area, the individual must be able to understand the functions, mechanical components and systems of the milling machine.

LEARNING TASKS	CONTENT
1. Describe milling equipment	• Reference manufacturers and equipment available • Reference brochures, videos, etc.
2. Describe the function of milling equipment	• Milling: Profiling • Reclaiming: total reclaiming of existing materials • Material injection • Stabilized base • Foamed asphalt • Reclaim, grade, pave
3. Describe major mechanical components	• Major components checklist • Identify and describe function from list of major components
4. Describe potential failures, symptoms and indicators of failure	• Symptoms: noise, vibration, smell, leaks, cracks • Respond appropriately: service required, immediate repair required, immediate removal from service

Line (GAC): H OPERATE MILLING AND RECLAIMING EQUIPMENT

Competency: H2 Identify milling and reclaiming safety procedures

Objectives

To be competent in this area, the individual must be able to:

- Describe how to work safely while working on and around a milling machine. This will include understanding key regulations, machine limitations, operating procedures and tasks and the use of PPE's.
- Comply with safety requirements at all times during this course.

LEARNING TASKS	CONTENT
1. Interpret and follow job plan	• Communication of plan ○ Path ○ Pattern ○ Depth ○ Passes
2. Demonstrate awareness of machine-specific safety	• Dust, pressure (machine malfunctions/ kickbacks) blind spots, fumes, hydraulic line, noise • Block outs
3. Describe Traffic Control requirements	• Number of TCP's required • Signage • Signals • Communication
4. Describe proper PPE	• Workers are responsible to provide general purpose gloves, safety footwear, hard hat - employers are responsible for providing all other safety items • Employers must ensure that PPE is properly worn when required, and is cleaned, inspected, maintained • Employers must ensure that workers are adequately trained: correct use, limitations, maintenance
5. Describe lockout procedures	• Fuel supply • Battery lockout, etc.
6. Describe how to secure equipment for safety	• Vandal proof take ancillary parts off • Park safety block elevated equipment, use parking brakes and chocks

Line (GAC): H OPERATE MILLING AND RECLAIMING EQUIPMENT

Competency: H3 Describe milling and reclaiming operator roles and responsibilities

Objectives

To be competent in this area, the individual must be able to understand the roles and responsibilities of each of the team members of a milling crew.

LEARNING TASKS	CONTENT
1. Describe the role of Milling Machine Operator	• Top end
2. Describe the role of ground men	• Operating depth of drum • Assume profile depth, quality of job • Keep trucks in line, laser guided controls, set controls • Team contolling function • Cooperation is key
3. Demonstrate hand signals communication	• Not standardized

Line (GAC): H OPERATE MILLING AND RECLAIMING EQUIPMENT

Competency: H4 Perform milling and reclaiming equipment maintenance

Objectives

To be competent in this area, the individual must be able to perform regular daily maintenance on the milling machine.

LEARNING TASKS	CONTENT
1. Service lubrication system (greasing)	• Identify and lube all points • Load grease gun • Note: point out all that employers differ, but the following guidelines are reasonable • 2-3 shots for small areas: fan bearings, small U joints, linkages, hinges, etc. • 8-10 shots for common pins and bushings • 20 shots where one nipple feeds large areas • Every pin must be showing excess grease • Daily for most points/weekly for some • Wipe off nipples before greasing to reduce abrasive particles entering pin/bushing
2. Service air intake system as required (air filters / pre cleaners)	• Restriction indicators • Pre cleaner service • Cleaning vs. replacement, check policy • Use air only with reduced pressure
3. Drain fuel tank sump, water separator fuel filters if equipped	• Drain, assess • Water, debris, scale, etc.
4. Add fluids as required	• Indetify correct fluids for each components • Add fluids as required • Avoid contamination with dirt, water • Describe 2 sided dipsticks with levels for stopped/idling (CAT engines, transmissions)
5. Inspect and clean components as required	• Radiation and oil cooler • Filter plugs, dipstick • Battery terminals
6. Perform housekeeping tasks	• Garbage • Excess grease • General cleanliness

Achievement Criteria

Performance	The learner will perform regular daily maintenance on the milling machine. Learner will demonstrate safe practices during practical training and on the job.
Conditions	The learner will demonstrate all elements of daily maintenance.
Criteria	The learner will demonstrate a minimum of 80% efficiency using a checklist.

Line (GAC): H OPERATE MILLING AND RECLAIMING EQUIPMENT

Competency: H5 Comply with markers, grades and stakes

Objectives

To be competent in this area, the individual must be able to describe and apply job control and basic engineering knowledge.

LEARNING TASKS	CONTENT
1. Describe abbreviations used on stakes	• Abbreviations commonly used on stakes ○ B/M - bench mark ○ C - cut ○ C/B - catch basin ○ SMH - sanitary manhole ○ Elv or EL - elevator ○ F/G - finish grade ○ P/L - property line ○ R/W - right of way ○ SAN or S - sanitary ○ Mon - monument ○ G - gas ○ PVC - poly vinyl chloride
2. Interpret stakes	• Interpret common types of staking ○ Property line, iron pins ○ Centre line, stationing ○ Offset ○ Cut, fill ○ Grade ○ Slope ○ Reference point ○ Bench marks, temporary bench marks: geodetic and referenced to assumed elevations ○ Survey menument: geodetic, precise, permanent, protected by law ○ Cross-heads ○ Location of work, e.g., manholes, catch, basins, headwalls, etc.

Achievement Criteria

Performance — Learner will demonstrate job control and basic engineering knowledge.
Learner will demonstrate safe practices during practical training and on the job.

Conditions — The learner will apply information from the stakes to determine

- C - cut
- C/B - catch basin
- SMH - sanitary manhole
- Elv or EL - elevator
- F/G - finish grade
- G - gas
- Centre line, stationing
- Offset
- Cut, fill
- Grade
- Slope

Criteria — The learner will be capable of maintaining

- C - cut
- C/B - catch basin
- SMH - sanitary manhole
- Elv or EL - elevator
- F/G - finish grade
- G - gas
- Centre line, stationing
- Offset
- Cut, fill
- Grade
- Slope

Line (GAC): H OPERATE MILLING AND RECLAIMING EQUIPMENT

Competency: H6 Operate milling and reclaiming equipment

Objectives

To be competent in this area, the individual must be able to perform basic operating functions during practical training.

LEARNING TASKS	CONTENT
1. Plan job	• Discuss the importance of proper planning • Follow plan for practical in-school training
2. Position the machine	• Manoeuvre machine into position to begin milling • Position haul truck for material collection
3. Engage teeth	• Lower cutting wheel
4. Move forward	• Follow the line (could be a string, paint, or previous cut)
5. Make adjustments	• Avoid adjustments if possible – when necessary do not make dramatic adjustments
6. Control depth	• Proper use of depth controls, set according to job requirements
7. Set Controls	• Proper speed, depth and angle for job requirements

Achievement Criteria

Performance: The learner will demonstrate basic operating functions.
Learner will demonstrate safe practices during practical training and on the job.

Conditions: To complete the in school practical and jobsite requirement for the milling machine, the learner must be able to demonstrate the following practical skills:

- Demonstrate the safe operation of equipment including observation of surroundings
- Perform pre-start checks, start-up/shut-down procedures and monitor performance of the equipment
- Perform daily maintenance tasks

Criteria

Perform basic moves with equipment including

- Move forward, stop, back up, stop (flat elevation)
- Apply park brake, lock out
- Raise, lower and swing conveyor (making sure to observe for wires)
- Set controls for proper depth of cut
- Mill a reasonable section of asphalt or compacted gravel to give trainee a “feel” for the equipment – site conditions will dictate actual extent of simulated milling
- Demonstrate effective communication, teamwork
- Follow the line (painted line or previous cut)

Communicate with haul truck driver (note: some training sites set up milling machine for side discharge with no haul truck present)

Line (GAC): H OPERATE MILLING AND RECLAIMING EQUIPMENT

Competency: H7 Follow shut down procedures

Objectives

To be competent in this area, the individual must be able to perform correct shut down procedures for the milling machine.

LEARNING TASKS	CONTENT
1. Perform shut down procedures	• Position that allows access in case of mechanical trouble at start up • Safe parking position, set parking brake • Level position for checking of fluids • Cool down before shut down • Walk around, look for leaks, general check secure manchine, locks, etc.

Achievement Criteria

Performance	The learner will demonstrate correct shut down procedures for the milling machine.
Conditions	The learner will follow shutdown procedure as listed by the manufacturers guidelines.
Criteria	The learner will demonstrate a minimum of 80% efficiency using a checklist

Line (GAC): H OPERATE MILLING AND RECLAIMING EQUIPMENT

Competency: H8 Transport milling and reclaiming equipment

Objectives

To be competent in this area, the individual must be able to understand the correct transport procedures for the milling machine.

LEARNING TASKS	CONTENT
1. Describe roles and responsibilities	• Truck driver is usually responsible for transporting equipment
2. Prepare machine for transport	• Remove attachments • Secure equipment • Ensure all guards are in place
3. Describe loading and unloading procedures	• Operator is usually involved in loading/unloading • Use of tie-downs • Safe movement of machine on/off trailer

Achievement Criteria

Performance	The learner will demonstrate safe practices while preparing machine for transport. Learner will demonstrate safe practices during practical training and on the job.
Conditions	The learner will prepare the machine for transport using appropriate tools and equipment to manufactures specification.
Criteria	The learner will be able to secure and prepare a machine for transport within a given time frame.

Line (GAC): I ASPHALT PAVER OPERATION

Competency: I1 Describe asphalt placement process

Objectives

To be competent in this area, the individual must be able to describe the processes of asphalt laydown from pre-laydown to compaction.

LEARNING TASKS	CONTENT
1. Describe how to place prime and tack coats	• When placing asphalt over existing asphalt, a tack coat is required
2. Describe where to position the Paver	• Asphalt train
3. Describe how to ready the Paving Machine	• Hydrostatic transmission and hydraulic systems require full power
4. Describe how to lower Screed on to blocks	• Proper thickness • Do not remove blocks until machine has moved forward • Safe movement of machine on/off trailer • Allow approx 20% for compaction
5. Describe how to place Haul Truck Loads in Hopper	• Communnicate with proper hand/horn signals • Open hopper wings
6. Describe how to place the asphalt product	• Joints, affective joints and overlaps • Maintain proper head of material on screed • Ensure appropriate speed
7. Describe how to compact asphalt	• Explain breakdown, vibration, compaction and the effects of temperature, etc.

Line (GAC): I ASPHALT PAVER OPERATION

Competency: I2 Describe paving machine equipment and attachments

Objectives

To be competent in this area, the individual must be able to describe the functions of each piece of equipment and attachment used in the paving process.

LEARNING TASKS	CONTENT
1. Describe types of paving machines	• 8 and 10ft paver (standard) • Light duty • Sidewalks, pathways, shoulders • Medium duty • Highway class • Specialty pavers
2. Describe major components	• Two parts ○ Tractor ○ Rubber tire, rubber tracked or steel tracked ○ Hopper handles material, pulls the screed ○ Screed (attachment) ○ Different widths, uses, etc. ○ Variable and fixed
3. Describe attachments	• Pick up machine • Picks up asphalt • Hopper system • Remixes it for consistent output • Alleviates segregation • Uses belly dump instead of end dump (more efficient)

Line (GAC): I ASPHALT PAVER OPERATION

Competency: I3 Apply paver work safety procedures

Objectives

To be competent in this area, the individual must be able to:

- Describe how to work safely while working on and around a paver. This will include understanding key regulations, machine limitations, operating procedures and tasks and the use of PPE's.
- Comply with safety requirements at all times during this program.

LEARNING TASKS	CONTENT
1. Interpret job plan	• Communicate job plan • Path • Pattern • Asphalt thickness • Crown • Obstacles
2. Demonstrate awareness of machine-specific safety	• Dust, pressure (machine malfunctions/kickbacks) blind spots, fumes, hydraulic lines, noise • Block outs • Pinch point hazard • Maintain safe distance from other equipment – e.g., Compaction Rollers
3. Describe traffic control requirements	• Signage • Signals • Communication
4. Use proper PPE	• Workers are responsible to provide general purpose gloves, safety footwear, hard hat - employers are responsible for providing all other safety items • Employers must ensure that PPE is properly worn when required, and is cleaned, inspected, maintained • Employers must ensure that workers are adequately trained: correct use, limitations, maintenance
5. Perform lockout procedures	• Fuel supply • Battery lockout, etc.

LEARNING TASKS	CONTENT
6. Secure machine for safety	• Vandal proof take ancillary parts off • Park safety block elevated equipment, use parking brakes and chocks

Achievement Criteria

Performance	The learner will demonstrate safe work practices while working with and operating the asphalt paver. Learner will demonstrate safe practices during practical training and on the job.
Conditions	The learner will: • Use correct PPE • Demonstrate machine specific safety awareness
Criteria	The learner will demonstrate correct lock-out procedures. Secure the machine for safety, including: • Chalks and blocks • Using night switch

Line (GAC): I ASPHALT PAVER OPERATION

Competency: I4 Asphalt paver operator roles and responsibilities

Objectives

To be competent in this area, the individual must be able to understand the roles and responsibilities of the asphalt paver crew team member.

LEARNING TASKS	CONTENT
1. Describe how to position Paver	• Efficient movement and alignment
2. Describe how to ready the Paver	• Heating the screed and on blocks
3. Describe how to follow the line	• Laser or string line guidance system
4. Describe the transition process	• Efficient loading of hopper
5. Describe how to maintain consistent “head”	• Keep hopper full • Material pushed by screed
6. Describe positioning of haul trucks	• Flow of mix from truck into hopper and hopper to the screed
7. Describe signals/communication	• Hand signals should be slow and controlled • Operator hand signals through truck mirrors and horn
8. Identify obstacles and overhead hazards	• Paver operator should be aware of obstacles (overhead and on path)